Next Generation Data Centers and Energy Management Systems

Lecture 7A: Overview of Simulation Field; the Concept of a Digital Twin

Prof. Thomas M. Schutzius

Department of Mechanical Engineering

Fall 2026

Berkeley
UNIVERSITY OF CALIFORNIA

Learning Plan

- Introduction to Data Center Systems and Thermodynamics
- Psychrometrics and Heat Transfer
- Performance Metrics and Standards
- Applied Computational Fluid Dynamics (CFD)
- Liquid Cooling & Advanced Heat Transfer
- Chip Cooling: Heat Conduction and Thermal Interface Materials
- Power Delivery and Infrastructure (Energy sources: Geothermal, Solar, Nuclear, etc.)
- **Overview of Simulation Field; the Concept of a Digital Twin**
- Machine Learning and Optimization Tools
- Modeling and Simulation of an Energy Management System
- Control Systems and Optimization
- Spacecraft Thermal Control
- Final Project Work, Peer Feedback, and Mentor Check-in
- Symposium

Learning Objectives

- Design a digital twin
- Create a reduced order model for a 2D transient heat conduction problem with heat generation

Typical layout of a data center facility

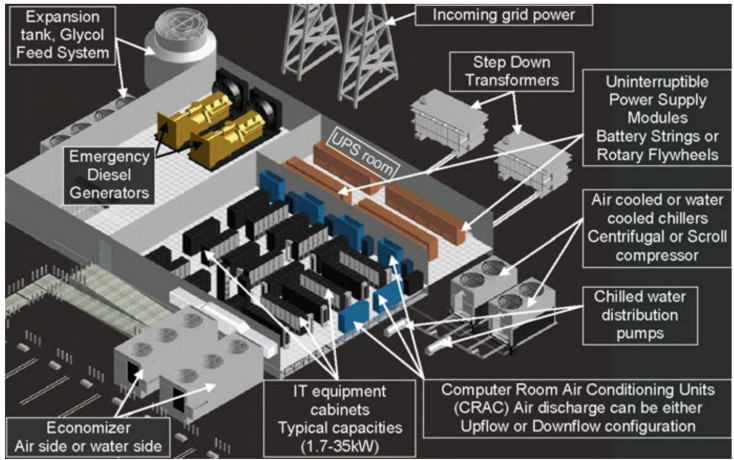


Figure 1: (Figure 1.1 in Ref. [1]).

Seminar 5

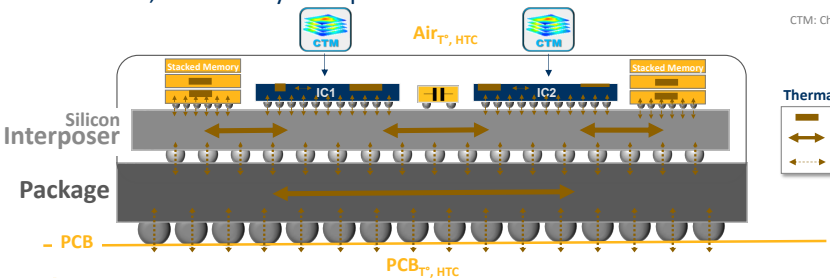
- Anjney Midha, AMP
- Date: October 22, 2026
- Time: 15:30 (Berkeley Time)
- Location: GSPP 150

Chip Thermal Modeling for 3DICs

- Predict the thermal hotspot of the die across multiple on-die routing layers
- Hotspot locations can guide on-die sensor placement and thermal management
- Chip/chiplet level static and transient temperature response
- RedHawk-SC Electrothermal is our chip-level thermal sign-off solution, certified by multiple foundries



CTM: Chip Thermal Model



Thermal Conductive Paths

sources	✓ IC Power Density
intrinsic	✓ Metal Densities
TSV, Bumps	✓ Materials
	✓ Boundary Conditions

© 2026 Synopsys, Inc. and/or Ansys, Inc. All Rights Reserved.

Ansys part of SYNOPSYS

Digital Twin Introduction

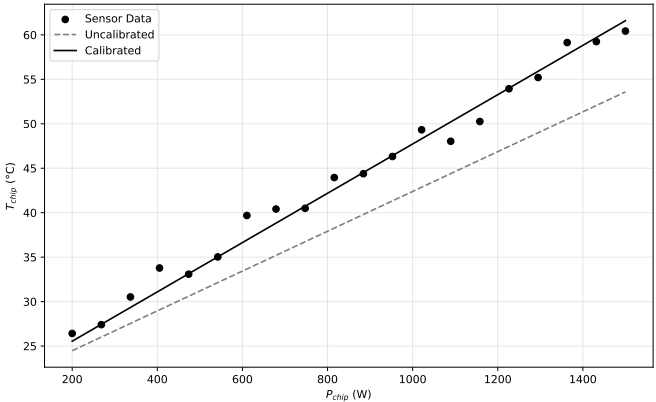


Figure 2: Calibrating a digital twin.

Reduced Order Modeling

References I

[1] Yogendra Joshi and Pramod Kumar, editors.
Energy Efficient Thermal Management of Data Centers.
Springer US, 2012.